Linear algebra – 3

Eigenvalues, diagonalisability and the Jordan form

Homework

12 сентября 2026 года

Theoretical minimum. A polynomial identity $f(A) = O$ puts the spectrum inside the roots of f, and if those roots are simple the matrix is diagonalisable; this is the shortest route in most problems. The Jordan structure is read off from the ranks of the powers of $A - \lambda E$, and the Frobenius form shows that similarity does not depend on the field extension. For a symmetric real matrix everything is orthogonal and the eigenvalues are real; the coefficients of χ_A are the elementary symmetric functions of the spectrum, so statements about $\sum \lambda_i$, $\sum_{i<j} \lambda_i \lambda_j$, $\prod \lambda_i$ are statements about traces, principal minors and the determinant. A matrix with constant column sums has that constant as an eigenvalue, with eigenvector $(1, \dots, 1)^{\mathsf{T}}$ of the transpose. Finally, if A is real and λ is a non-real eigenvalue, then $\overline{\lambda}$ is one too, with the conjugate eigenvector — half the problems about real matrices are solved by this remark alone.

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1. (IMC 2020) Let n be a positive integer. Find all $n \times n$ real matrices A with only real eigenvalues satisfying $A + A^k = A^{\mathsf{T}}$ for some integer $k \geqslant n$.

2. (IMC 2021) Let A be a real $n \times n$ matrix and suppose that for every positive integer m there exists a real symmetric matrix B with $2021B = A^m + B^2$. Prove that $|\det A| \leqslant 1$.

3. (IMC 2017) Determine all complex numbers λ for which there exist a positive integer n and a real $n \times n$ matrix A such that $A^2 = A^{\mathsf{T}}$ and λ is an eigenvalue of A.

4. (IMC 2025) For $A \in M_n(\mathbb{R})$ denote by A^R its counter-clockwise rotation by $\frac{\pi}{2}$, so that the first row of A^R is the last column of A read downwards,

and so on. Prove that if $A = A^R$, then every eigenvalue λ of A satisfies $\operatorname{Re}\lambda = 0$ or $\operatorname{Im}\lambda = 0$.

5. (IMC 2003) Let A be an $n \times n$ real matrix such that $3A^3 = A^2 + A + E$. Show that the sequence A^k converges to an idempotent matrix.

6. (IMC 1997) Let A and B be real $n \times n$ matrices such that $A^2 + B^2 = AB$. Prove that if $BA - AB$ is an invertible matrix then n is divisible by 3.

7. (IMC 1999)

   a) Show that for every $m \in \mathbb{N}$ there exists a real $m \times m$ matrix A with $A^3 = A + E$.

   b) Show that $\det A > 0$ for every real $m \times m$ matrix satisfying $A^3 = A + E$.

8. (IMC 1996) A linear operator A on a vector space V is called an involution if $A^2 = E$. Let $\dim V = n < \infty$.

   a) Prove that for every involution A on V there is a basis of V consisting of eigenvectors of A.

   b) Find the maximal number of distinct pairwise commuting involutions on V.

9. (IMC 2018) Determine all rational numbers a for which the matrix

$$\begin{pmatrix} a & -a & -1 & 0 \\ a & -a & 0 & -1 \\ 1 & 0 & a & -a \\ 0 & 1 & a & -a \end{pmatrix}$$

   is the square of a matrix with all rational entries.

10. (VJIMC 2025) Let $n \geqslant 2$ be an integer, and let A be an $n \times n$ real matrix with minimal polynomial $x^n + x - 1$. Show that

$$\operatorname{tr}\big((nA^{n-1} + E)^{-1} A^{n-2}\big) = 0.$$

11. (Romania national olympiad 2018) Let $n > 2$ and let $A \in M_n(\mathbb{C})$ satisfy $\operatorname{rank} A \neq \operatorname{rank} A^2$. Prove that there is a non-zero $B \in M_n(\mathbb{C})$ with $AB = BA = B^2 = O$.

12. (IMC) Let $A \in M_n(\mathbb{C})$ with $A \neq \lambda E$ for every $\lambda \in \mathbb{C}$. Prove that A is similar to a matrix having at most one non-zero entry on the main diagonal.

13. (IMC) Let A, B be real $n \times n$ matrices and suppose there are $n+1$ distinct real numbers $t_1, \ldots, t_{n+1}$ such that every $C_i = A + t_i B$ is nilpotent. Show that both A and B are nilpotent.

14. (IMC) Let $A = (a_{k\ell})$ be a complex $n \times n$ matrix such that for every m and all $1 \leqslant j_1 < \cdots < j_m \leqslant n$ the determinant of $(a_{j_k j_\ell})_{k,\ell=1}^m$ is zero. Prove that $A^n = O$ and that some permutation σ makes $(a_{\sigma(k)\sigma(\ell)})$ strictly upper triangular.

15. (SEEMOUS 2016)

    a) Prove that for every $X \in M_2(\mathbb{C})$ there is $Y \in M_2(\mathbb{C})$ with $Y^3 = X^2$.

    b) Prove that there is $A \in M_3(\mathbb{C})$ such that $Z^3 \neq A^2$ for every $Z \in M_3(\mathbb{C})$.

16. (IMC 2026) Let $n \geqslant 5$, and suppose that $A = (a_{ij})$ is a real symmetric $n \times n$ matrix such that $a_{ii} = 0$ and $a_{ij} \in \{-1, 1\}$ for $i \neq j$. Assume that the scalar products of any two distinct rows of A have the same value. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A. Prove that

$$\sum_{i=1}^{n} |\lambda_i| \geqslant 2n - 2$$

    and determine all matrices for which equality holds.